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(54) **USED PICKLEBALL PADDLE EVALUATION**

(71) Applicant: **Ofinno, LLC**, Reston, VA (US)

(72) Inventor: **Esmael Hejazi DINAN**, McLean, VA (US)

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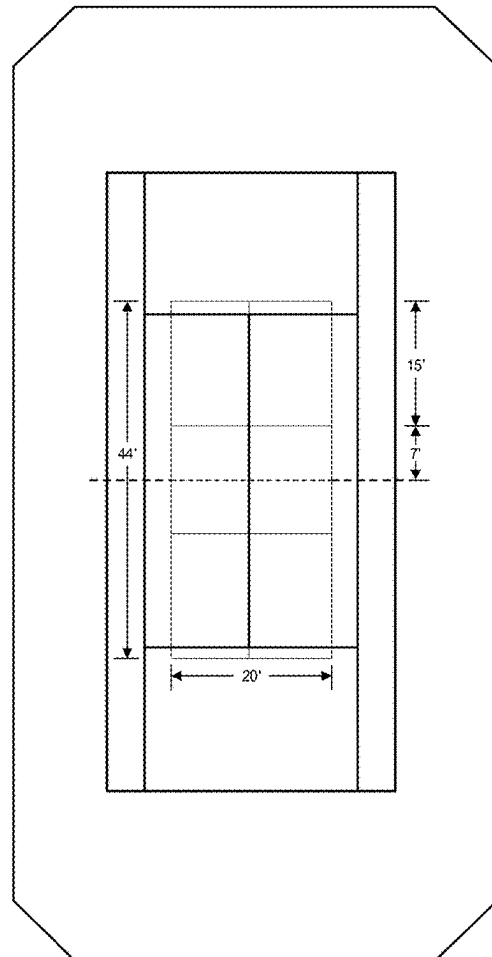
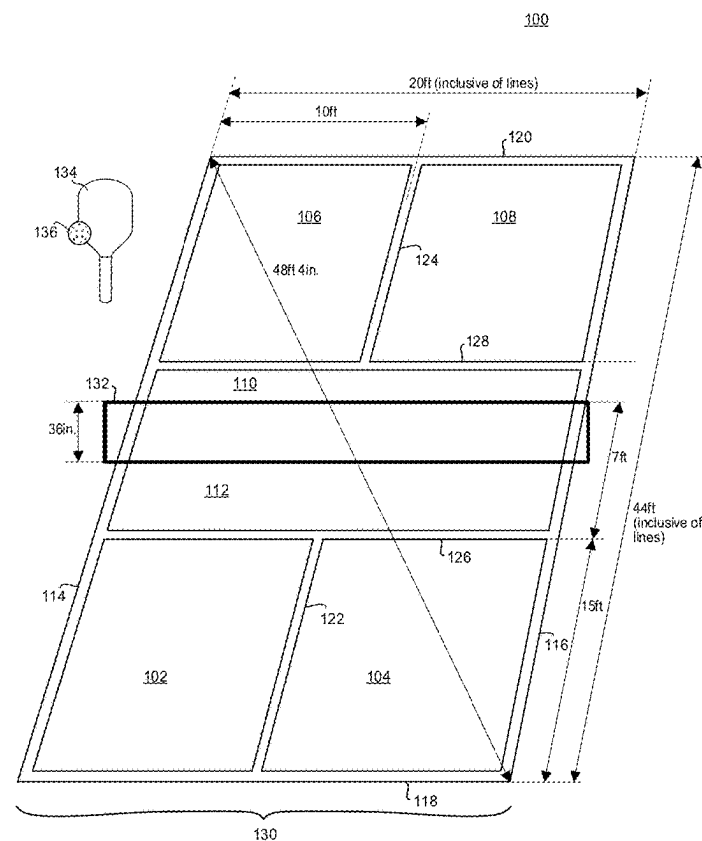
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(57)

ABSTRACT

Apparatuses and methods for evaluating pickleball paddles, such as, for example, used pickleball paddles, are described. An example pickleball paddle comprises a striking blade, a handle connected to the striking blade, and an electronic component including a memory storing one or more parameters associated with how the pickleball paddle is used after the pickleball paddle is manufactured. Another example pickleball paddle comprises an electronic component including a memory storing one or more parameters associated with how the pickleball paddle is used after the pickleball paddle is manufactured and where the electronic component is connected to an electronic display. An example evaluation process for testing a pickleball paddle is also described.



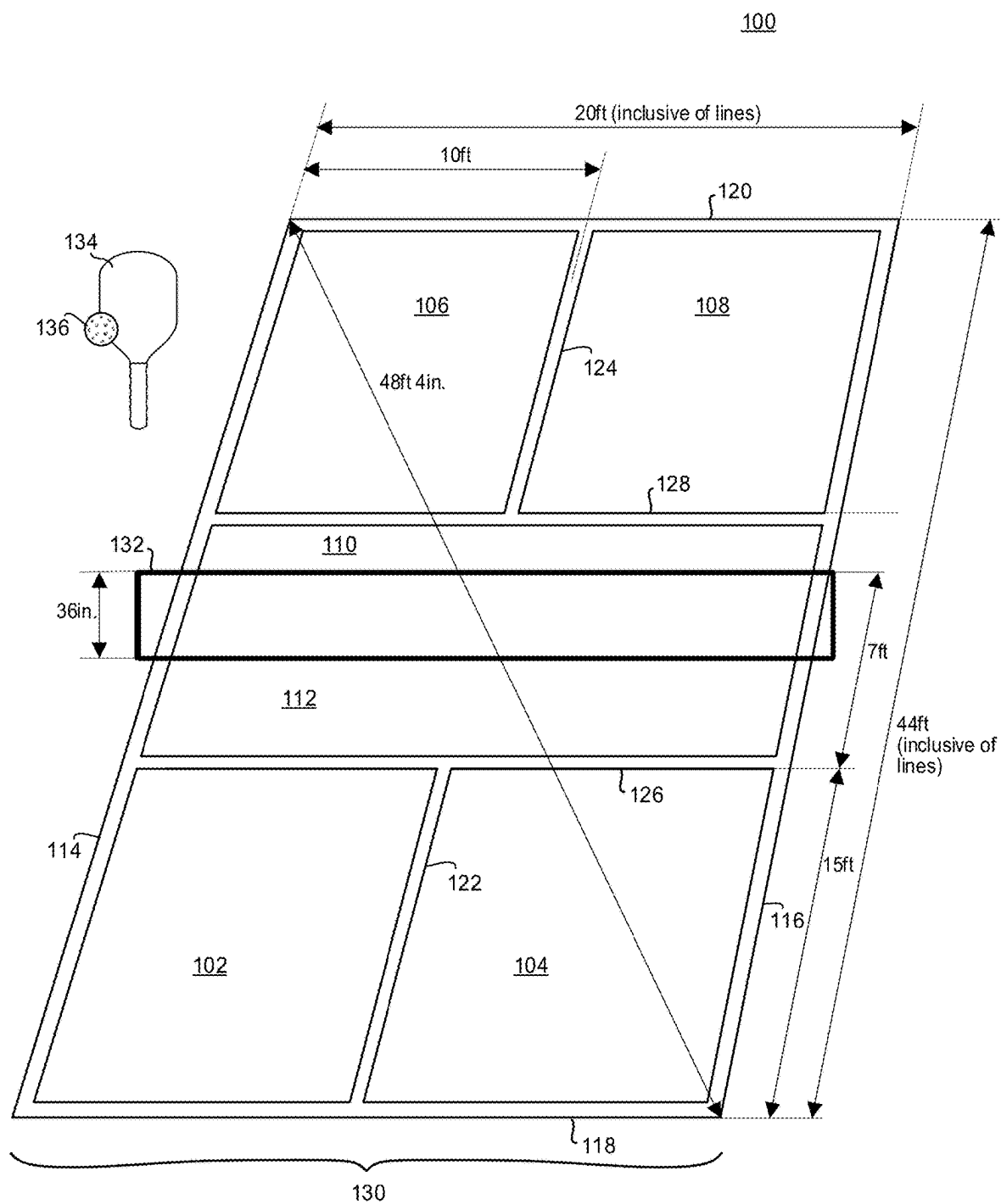


FIG. 1A

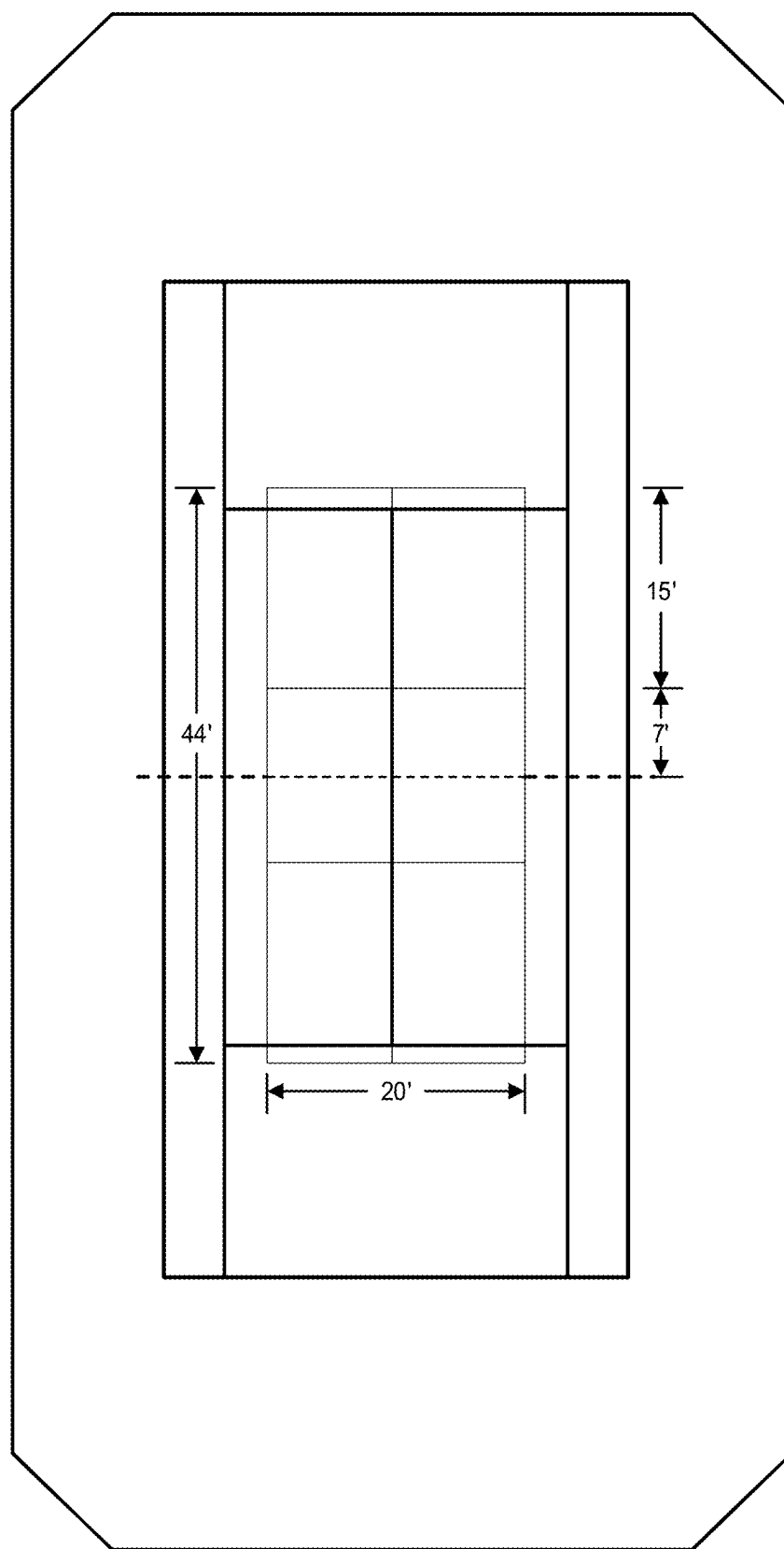


FIG. 1B

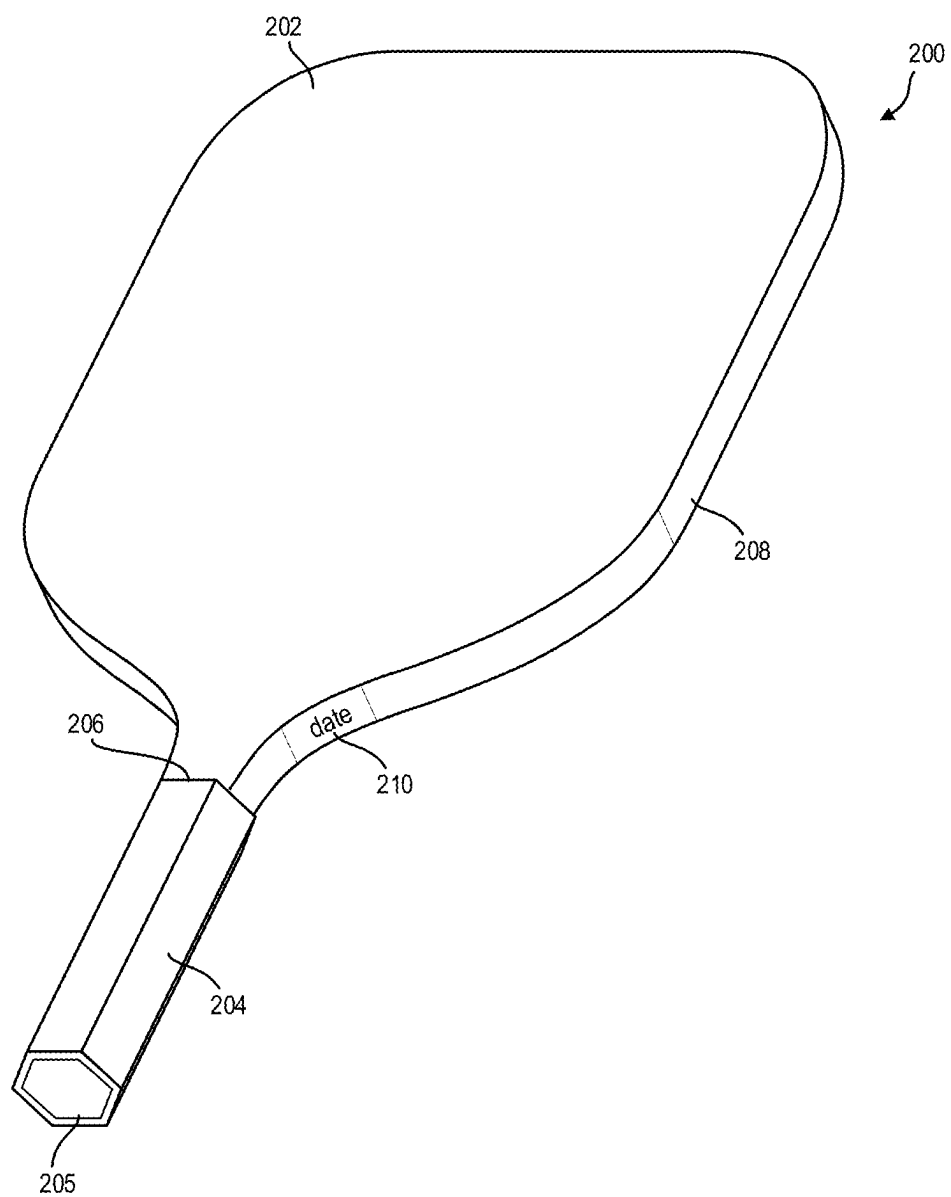


FIG. 2

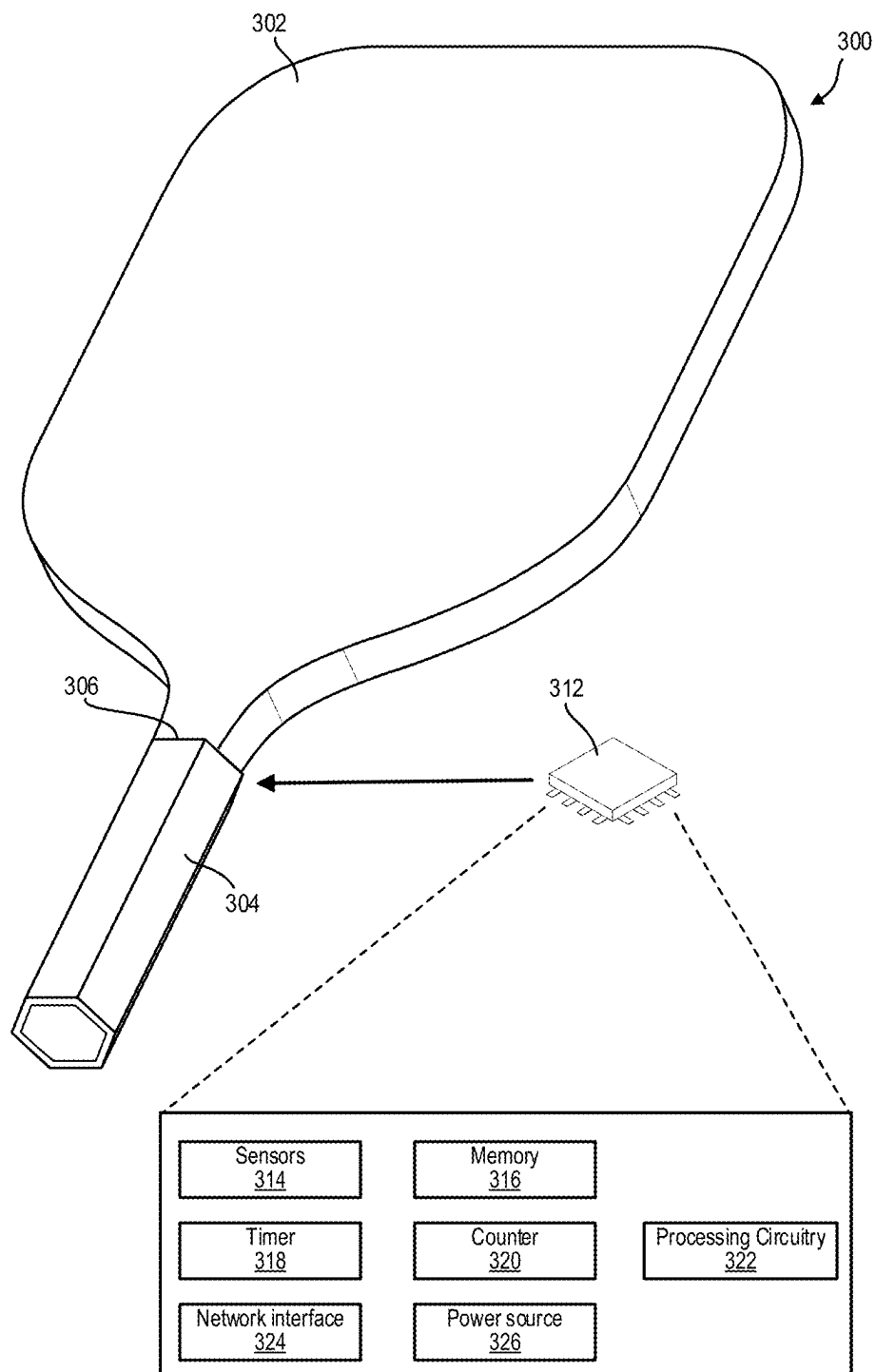


FIG. 3

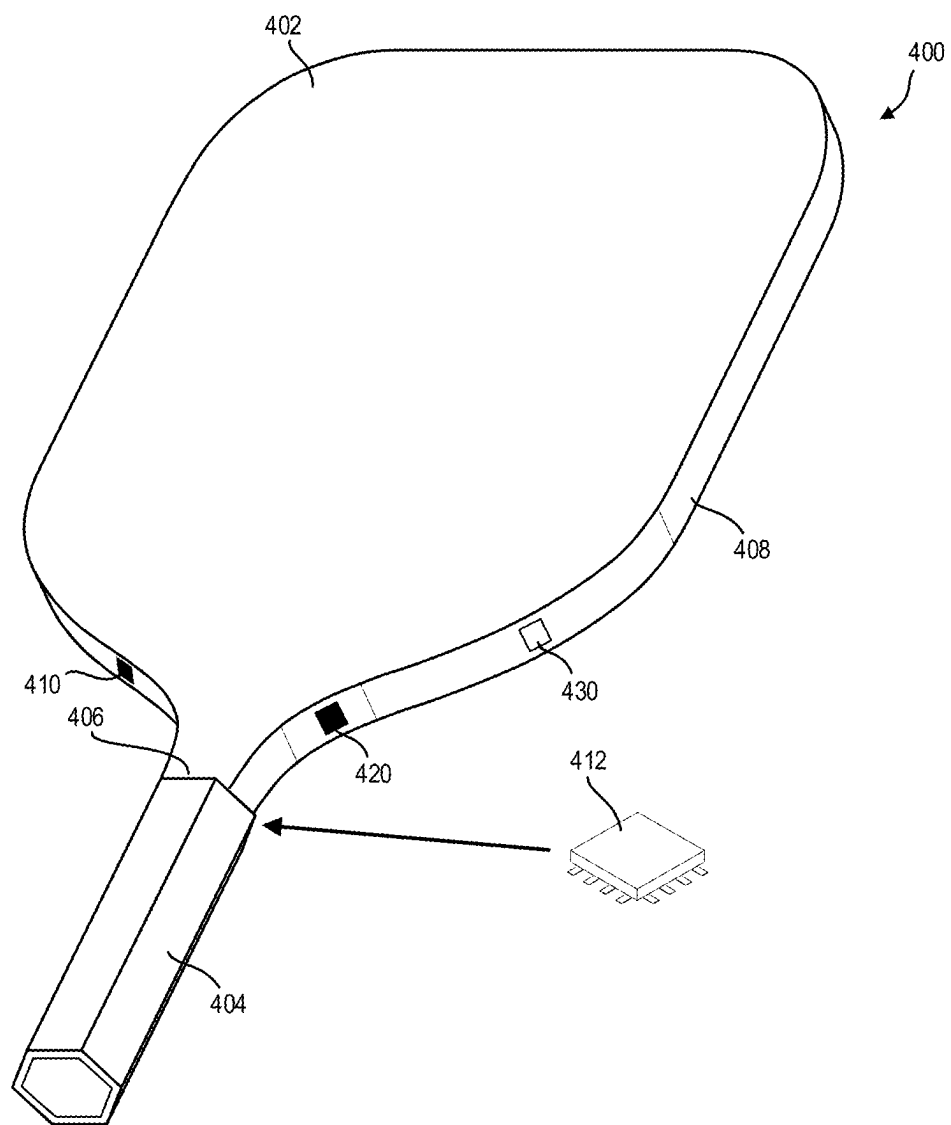


FIG. 4

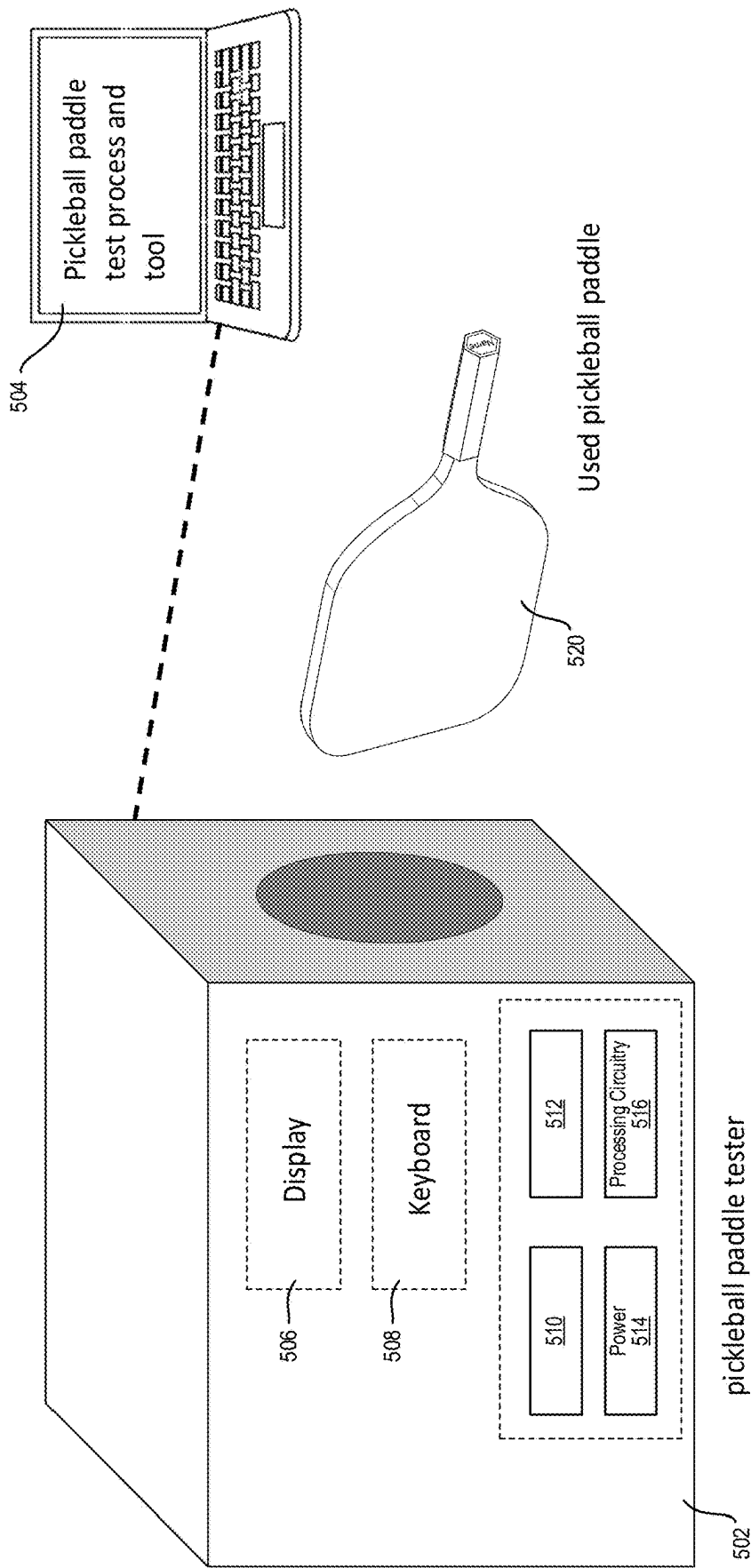


FIG. 5

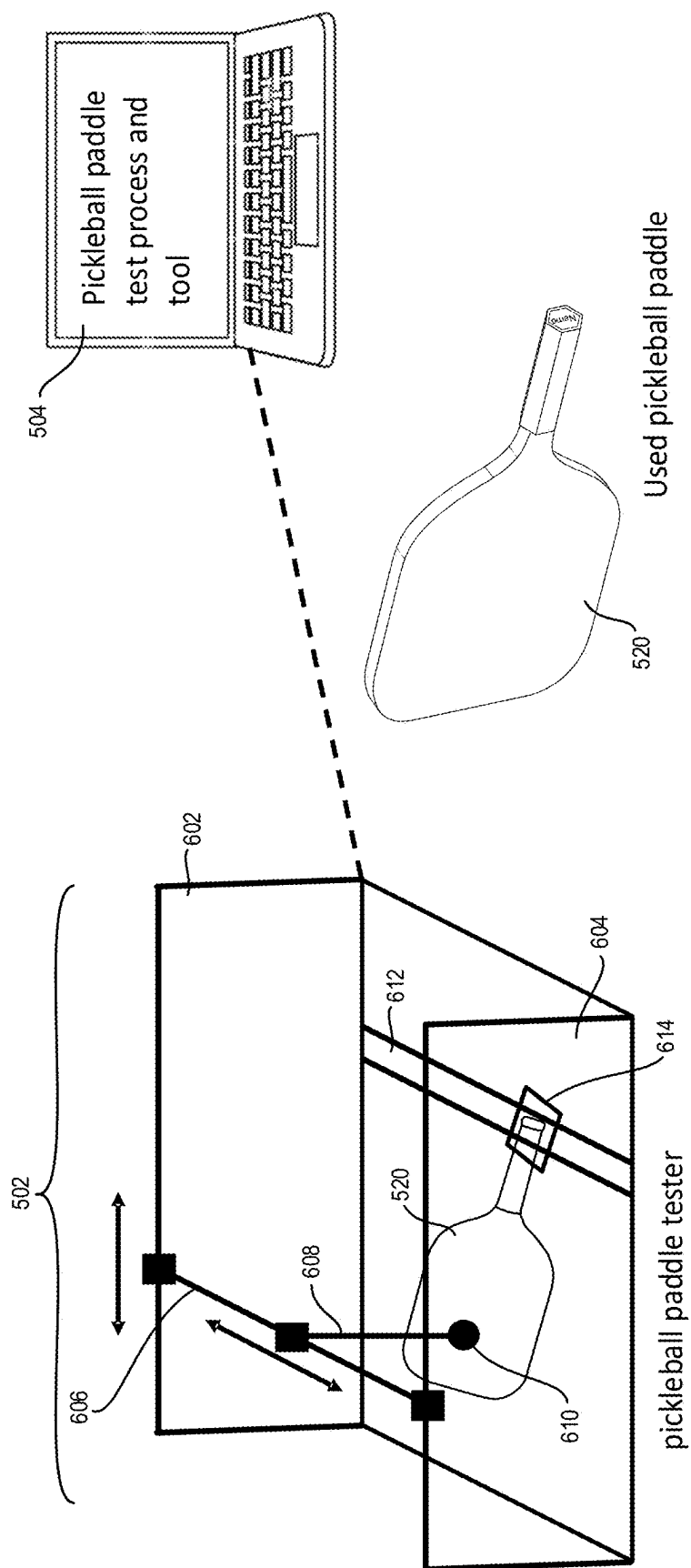
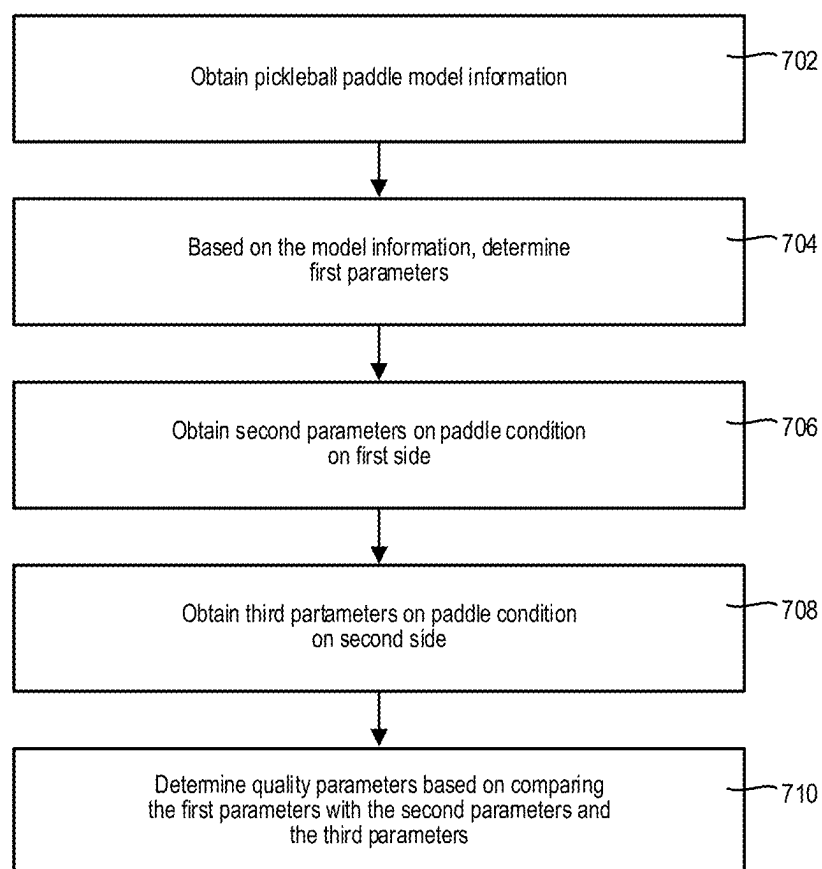


FIG. 6

**FIG. 7**

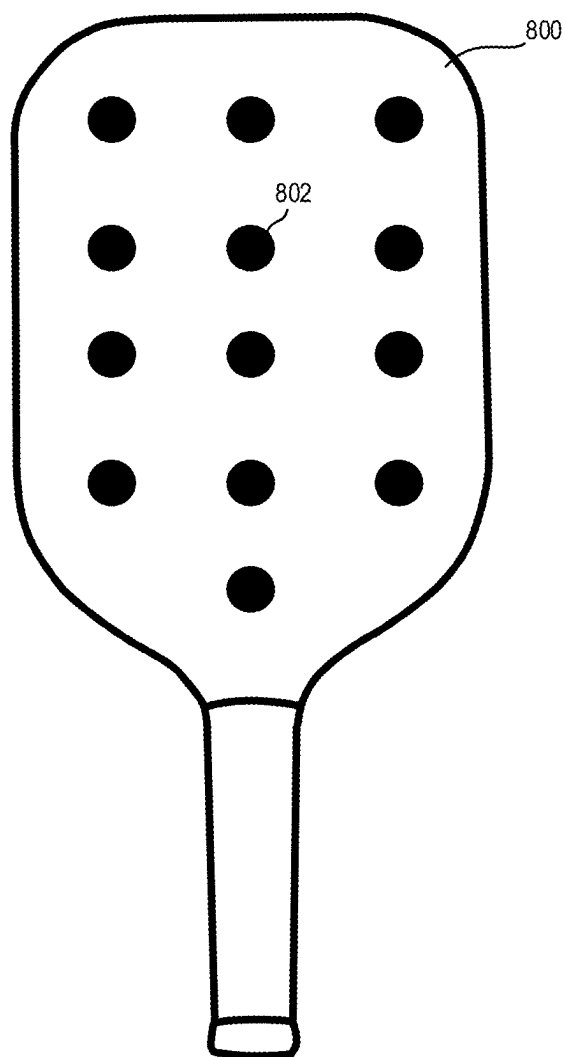


FIG. 8

Measurement Point	Side 1 – used paddle	Side 1 – New paddle	Difference	Percentage
1	7	7	0	0
2	6	6	0	0
3	5	8	3	12.5%
4	7	10	3	30%
5	7	10	3	30%
6	7	10	3	30%
7	10	12	2	16.6%
8	10	12	2	16.6%
9	6	12	6	50%
10	12	14	2	14.28%
11	12	14	2	14.28%
12	8	7	-1	-14.28%
13	8	8	0	0

FIG. 9

USED PICKLEBALL PADDLE EVALUATION

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to U.S. Provisional Application No. 63/558,108 filed Feb. 26, 2024, the entire content of which is herein incorporated by reference.

FIELD

[0002] This disclosure related to pickleball paddles, and in particular to the evaluation of pickleball paddles.

BACKGROUND

[0003] Pickleball is a paddle sport. Pickleball paddles are relatively expensive. The prices may range between \$50 up to about \$300 for high end paddles. Different paddles are made for different playing styles. For example, different pickleball paddles introduce different levels of control, power or spin on a pickleball when hitting the pickleball during a game.

[0004] A damaged pickleball paddle may deteriorate a player's game. For example, it may result in inconsistent serves, not producing enough power in drives, inaccurate dinking and shots, etc. Most damaged pickleball paddles may not easily be recognizable because the damage may not be visible. In other sports like tennis and racquetball, the string of a damaged racquet can be replaced. However, most damaged pickleball paddles are not repairable and should be discarded. It is estimated that millions of damaged pickleball paddles are owned by players and most of those players are not aware of it.

[0005] There is a need for a method and apparatus to identify damaged pickleball paddles.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Some features are shown by way of example, and not by limitation, in the accompanying drawings. In the drawings, like numerals may reference similar elements.

[0007] FIG. 1A shows an example pickleball court, in which embodiments of the present disclosure may be applied.

[0008] FIG. 1B shows another view of a pickleball court.

[0009] FIG. 2 shows a pickleball paddle configured to facilitate used pickleball paddle evaluation in accordance with some embodiments of the present disclosure.

[0010] FIG. 3 shows a pickleball paddle including an electronic component configured to facilitate used pickleball paddle evaluation, in accordance with some embodiments of the present disclosure.

[0011] FIG. 4 shows a pickleball paddle including an electronic component and a display element configured to facilitate used pickleball paddle evaluation, in accordance with some embodiments of the present disclosure.

[0012] FIG. 5 shows an example used pickleball paddle testing device and an optional connected computer, according to some embodiments of the present disclosure.

[0013] FIG. 6 shows an example test mechanism that may be configured in the test device shown in FIG. 5, according to some embodiments of the present disclosure.

[0014] FIG. 7 shows a flowchart of a process for testing a used pickleball paddle in a testing device such as that shown in FIG. 5, according to some embodiments of the present disclosure.

[0015] FIG. 8 shows an example set of test locations on the used pickleball paddle that may be tested by a test device such as the test device shown in FIG. 5, according to some embodiments of the present disclosure.

[0016] FIG. 9 shows an example set of results obtained from the process shown in the flowchart of FIG. 7, according to some embodiments of the present disclosure.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

[0017] In the following description, numerous specific details are set forth in order to provide a thorough understanding of the disclosure. However, it will be apparent to those skilled in the art that the disclosure, including structures, systems, and methods, may be practiced without these specific details. The description and representation herein are the common means used by those experienced or skilled in the art to most effectively convey the substance of their work to others skilled in the art. In other instances, well-known methods, procedures, components, and circuitry have not been described in detail to avoid unnecessarily obscuring aspects of the disclosure.

[0018] References in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described.

[0019] This disclosure relates to “Pickleball,” a game that has in recent years seen a massive increase in popularity among all age groups of players, including among senior citizens. Pickleball is a game that can be described as combining aspects of tennis, badminton, and ping-pong. It is played on a badminton-sized court, with paddles and a ball similar to a wiffle ball, but slightly smaller in size. The net used in pickleball is similar to a tennis net in some ways, but is lowered at the center.

[0020] Pickleball has recently become very popular and is played both indoors and outdoors, as either doubles or singles. The rules are relatively simple and the game is easy for beginners to learn. However, among skilled players, pickleball can develop into a quick, fast-paced, and competitive game.

[0021] FIG. 1A shows an example pickleball court 100 that includes a pickleball playing surface 130 and a pickleball net 132. The playing surface 130 comprises two left serve areas 102 and 108, two right serve areas 104 and 106, and two non-volley areas 110 and 112, with one of the left serve areas, one of the right serve areas, and one of the non-volley areas being on each side of the pickleball net 132. The net is 36 inches tall at the edges, and lowered to 34 inches in the middle. The areas 102-112 are defined by baselines 118 and 120 each 20 feet, sidelines 114 and 116 each 44 feet, center lines 122 and 124 each 15 feet, and non-volley lines 126 and 128 each the same size as a baseline. Each of the lines may be 2 inches wide. The term “court line” is used in this disclosure to refer to any sideline, baseline, centerline or non-volley line on the pickleball

court. Non-volley areas **110** and **112**, each extending 7 feet from the net, are also referred to as the “kitchen”.

[0022] The game of pickleball is played with a pickleball paddle **134** and pickleball ball **136**. The ball **136** is typically made of plastic and has a 3-inch diameter. Similar to a wiffleball, the ball **136** has through holes throughout the surface. Different types (e.g., with different levels of hardness and different sizes of the through holes) may be used for playing the game on the various types of pickleball courts (e.g., indoor, outdoor, hard surface, soft surface etc.).

[0023] Pickleball can be played as singles or doubles, and is most commonly played as doubles. Each point begins with an underarm serve. The serve is performed diagonally beginning at the right-hand service square. A valid serve sends the ball from one left serve area to the other left serve area or from one right serve area to the other right serve area. The serve must clear the non-volley-zone. The serve must bounce before being hit by the receiver. The return of serve must also bounce before being hit (this is known as the 2 bounce rule). After the serve and the return of the serve, the ball can land anywhere on the opposite side of the playing surface **130**. Volleys can only be performed outside of the non-volley zone. Volleys, that is, hitting the ball in the air without first letting it bounce, can only be made after the 2 bounce rule has been followed. However, if the ball is hit from within the kitchen, then it cannot land in the kitchen on the other side of the net.

[0024] A fault is any action that stops play due to a rule violation. A fault by the receiving team results in the servers earning a point. A fault by the serving team results in the server's loss of service and/or side out. A fault is committed when the serve touches any part of the non-volley zone (including the line) or the ball is hit out of bounds.

[0025] Pickleball games are typically played without a referee and are self-judged. Each player makes the line calls as to whether the ball is in or out when the ball contacts the playing surface on that player's side. The game continues to at least 11 points and requires a 2-point difference for a win. FIG. 1B shows another view of a pickleball court.

[0026] Pickleball is a paddle sport. Two or four players use pickleball paddles typically made of composite materials to hit a pickleball. According to the USA Pickleball Equipment & Evaluation Committee (EEC), 1,185 new pickleball paddles were tested and approved in 2023. According to the 2023 APP Tour's Pickleball Participation Report, there are 36.5 million pickleball players in the United States. The global pickleball paddles market size is estimated to be worth USD 148.5 million in 2021 and is forecast to a readjusted size of USD 253.8 million by 2028 with a CAGR of 7.8% during the forecast 2022-2028.

[0027] Currently, it is estimated that there are tens of millions of pickleball paddles sold by paddle manufacturers. The pickleball paddles are relatively expensive. The prices may range between \$50 up to about \$300 for high end paddles. Different paddles are made for different playing styles. For example, different pickleball paddles introduce different levels of control, power or spin on a pickleball when hitting the pickleball during a game.

[0028] There are many different processes for manufacturing a pickleball paddle. For example, a core part of the pickleball paddle may be made of a polymer core with hexagon structure, which may be a lightweight and soft plastic-like material. A thin layer of carbon fiber, graphite, or other very thin layers of material may cover the polymer

core using an adhesive material. Polymer core may begin to deaden over time with repetitive impact. The thin cover layer may start to lose performance after tens of thousands of hits. The adhesive material that connects the cover to polymer core may lose its properties over time.

[0029] A damaged pickleball paddle may deteriorate a player's game. For example, it may result in inconsistent serves. A damaged pickleball paddle may not produce enough power in drives and may result in a ball hitting the net. A damaged pickleball paddle may result in inaccurate dinking and shots. Most damaged pickleball paddles may not easily be recognizable as the damage may not be visible. In other sports like tennis and racquetball, the string of a damaged racquet can be replaced. However, most damaged pickleball paddles are not repairable and should be thrown away. It is estimated that millions of damaged pickleball paddles are owned by players and most of those players are not aware of it. There is a need for a method and apparatus to identify damaged pickleball paddles. Some example embodiments provide methods and apparatus that enable determining if a pickleball paddle may not perform its functions. Those pickleball paddles may be discarded by their owners.

[0030] Pickleball paddles have a limited lifespan. The duration a pickleball paddle lasts may depend on how the paddle is taken care of and how frequently it is used and how it is used. Taking proper care of a pickleball paddle may extend the lifespan of the pickleball paddle. There are various ways of taking care of a pickleball paddle. For example, a pickleball paddle may be cleaned using a glass cleaner and/or a lint-free cloth, when it needs cleaning. Leaving a pickleball paddle in a hot or cold condition for a relatively long period of time may negatively affect its performance and lifespan. Keeping a pickleball paddle in a pickleball paddle case may increase its lifespan. Pickleball paddle cases may reduce the likelihood of damage due to impact or scratches. Hitting a pickleball paddle on hard materials may damage its structure.

[0031] For example, a pickleball paddle lifespan may be shortened if a player uses the pickleball paddle aggressively. Recreational players with a soft gaming style may get the most longevity out of a pickleball paddle. Normal wear and tear on a pickleball paddle may be accelerated by frequent use and aggressive play. Some players may reduce celebratory and/or post-game clinking the pickleball paddles to reduce the risk of additional damage. Leaning on a paddle, hitting a pickleball paddle on other hard objects may increase stress on the pickleball paddle and reduce its performance.

[0032] In general, a pickleball paddle may last a few months to a few years depending on various factors. Scratches, scrapes, and dead spots may be inevitable as a pickleball paddle is used over time. The issues in a pickleball paddle may be introduced in various parts of a paddle, for example in or around the sweet spot of a pickleball paddle, around the edges of a pickleball paddle, or on a handle of a pickleball paddle.

[0033] One important issue is that it is very hard to determine whether a used pickleball paddle is performing its desired functions effectively. It is generally tough to determine gradual and subtle issues in a pickleball paddle as they are not visible or easily detectable. A pickleball paddle may be subject to dead spots, internal fractures, and just overall wear and tear that may affect its performance. Many issues

on a pickleball paddle are not visible. A compromised pickleball paddle may negatively affect a player's performance. Most people determine whether a pickleball paddle is performing its functions based on how it feels when they use them.

[0034] Players often wonder whether it is time to change their pickleball paddles. They may use a pickleball paddle much longer than its lifespan. They may not recognize that they are losing games because their pickleball paddles are not performing as they were supposed to. This is also an issue for clubs that maintain a pool of pickleball paddles for players. They often carry a mix of damaged and proper pickleball paddles that are practically inseparable from each other.

[0035] The market for selling and buying used pickleball paddles is growing. Some pickleball paddles are relatively expensive and may change hands multiple times as some players decide to switch to a different brand or type of pickleball paddle. As the market for trading used pickleball paddles is growing, another issue appears to be critical. Currently, there is no mechanism to determine whether a used pickleball paddle is performing its functions properly or under-performs its functions.

[0036] There are various ways to measure various performance metrics of a pickleball paddle. For example, USA Pickleball Association (USAPA) developed a process for pickleball paddle testing. USAPA defines specifications for certifying a pickleball paddle. These specifications may include limits on a size, a roughness of the surface, a sheen/reflection factor, frictional characteristics, and a deflection (flexibility and response) parameter of pickleball paddles. The certification testing is developed for certifying pickleball paddles.

[0037] Unfortunately, the certification testing process may not be used to determine whether a paddle is performing well or not. A damaged pickleball paddle may successfully pass certification testing. For example, a used pickleball paddle that includes dead spots may successfully pass the tests related to deflection testing during pickleball paddle certification. The testing processes by USAPA are typically developed to ensure that some players do not take advantage of paddles that provide extra advantage during a game and are not developed to determine underperforming pickleball paddles. The certification testing may be impractical and inaccurate for determining whether a paddle is damaged considering that thousands of different paddle types are currently in the market. For example, a damaged high end pickleball paddle may perform better than a new low end pickleball paddle.

[0038] There is a need for apparatus, methods and systems to determine whether a pickleball paddle satisfactorily performs its functions. Some example embodiments may enable or support a user to determine whether a pickleball paddle functions properly and whether it is time to replace a pickleball paddle. Some example embodiments may enable a user to determine whether a used pickleball paddle performs according to its desired performance metrics. Depending on the complexity of an example apparatus, methods and systems, the accuracy of outcome may vary.

[0039] In some example embodiments, as shown for example in FIG. 2, a pickleball paddle 200 may comprise a striking blade 202 and a handle 204 connected to the striking blade. A grip 206 may wrap around a portion of the handle 204. A date 210 may be printed on the striking blade or the

handle. The date may be based on when the pickleball paddle is manufactured. In an example, the date may be printed on a surface of the striking blade. In an example, the date may be printed on a side 208 of the striking blade. In an example, the date may be printed on the handle. In an example, the date may be printed on a butt cap 205 connected to the handle. In an example, the date may comprise a year. In an example, the date may comprise a year the pickleball paddle is manufactured. In an example, the date may comprise a year model of the pickleball paddle. In an example, the date may comprise a year and a month. In an example, the date may comprise a year and a month the pickleball paddle is manufactured. In an example, the date may comprise a year and/or a month model of the pickleball paddle. In an example, the date may further comprise a day. In an example, the date on the paddle may be an indicator indicating the date. For example, the date may be coded in a form of number, a letter, or an alphanumeric code and be printed on the paddle. The form may indicate the date. In an example, the date may be attached to the paddle using a semi-permanent sticker.

[0040] A user may use the date to determine how old the pickleball paddle is. A user may determine the age of the pickleball paddle. A user may use the age to determine how used the paddle is, or how much the paddle is deteriorated. A user may use additional factors, for example, how frequently and/or how a paddle was used along with the age to determine a performance and/or quality of a paddle. For example, if a pickleball paddle was made 5 years ago, a user may determine that the paddle may not function according to its specifications and/or may not have a desired quality. For example, if a pickleball paddle was made 2 years ago and/or was used frequently may not have the desired quality.

[0041] A manufacturing date is typically printed on food products and/or sophisticated machinery. For example, a user may readily determine when certain packaged food was made, or when a vehicle was manufactured. Such a process has not been applied on pickleball paddles mainly because pickleball paddles were considered relatively simple products. Another reason many manufacturers do not print a date on a pickleball paddle may be because it is thought that a pickleball paddle may last for a substantially very long time, a deteriorated/damaged paddle may be identified readily and there is no need for printing a date, and/or a pickleball paddle is relatively cheap, and/or the like. As described in the specifications, printing a date on a pickleball paddle may enable a user to determine a quality of a paddle and whether a paddle meets a desired performance. The date may be used along with other information to determine a desired performance of a pickleball paddle and/or a pickleball paddle may be discarded. A user may use this information to determine whether it is time to replace a paddle. A user may use this information to determine whether to trade a used paddle and/or a price for trading the pickleball paddle.

[0042] In some example embodiments, as shown in FIG. 3, a pickleball paddle 300 may comprise a striking blade 302 and a handle 304 (optionally with a grip 306 wrapped around) connected to the striking blade, and an electronic component 312. The electronic component may include a memory 316 storing one or more parameters associated with how the pickleball paddle is used, e.g., after the pickleball paddle is manufactured. In an example embodiment, the one or more parameters is initialized to an initial value when the pickleball paddle is manufactured.

[0043] In some example embodiments, the electronic chip may be installed in the handle or in the striking blade. The one or more parameters may be updated as the pickleball paddle is used and/or how it is used. For example, at least one shock and/or movement sensor connected to the electronic chip may enable the electronic component to count approximately how many times the pickleball paddle hits a pickleball. The sensor and/or the count are expected to have some errors as the pickleball paddle may be moved without hitting a pickleball. The electronic component may enable determining within a predetermined accuracy how much the pickleball paddle is used. In an example, the at least one sensor may determine an intensity related to a movement of the pickleball paddle, for example, when the pickleball paddle hits a pickleball.

[0044] The counter may be incremented when the pickleball paddle movement changes direction as it hits a pickleball. In an example, the counter may be incremented by a value that is depending on how hard the pickleball paddle hits a pickleball. The sensor may be an impact sensor, a shock sensor, a sound sensor, acceleration sensor and/or the like (e.g., a piezoelectric sensor **314**). In an example embodiment, a value of the one or more parameters may depend on how frequently and how intensely the pickleball paddle was used.

[0045] In an example, the electronic component may include a timer **318** determining an age of the pickleball paddle. The electronic component may store a date that is based on when the pickleball paddle was manufactured. This information along with one or more parameters may be used to determine how a pickleball paddle is used and/or its performance. In an example, the one or more parameters may be based on an age of the pickleball paddle.

[0046] The electronic component may include or may be connected to one or more sensors **314**. The one or more sensors may be like sensors used in other movement sensors used in step sensors and/or may include accelerometer or a device that measures vibration or acceleration of motion of a pickleball paddle. In an example, a force caused by vibration or a change in motion (acceleration) may cause a mass in a sensor and/or electronic component to squeeze a piezoelectric material in the sensor. This may produce an electrical charge that is proportional to a force exerted upon it. The electrical charge may cause a counter or a parameter to change accordingly. The one or more parameters may store an accumulated value based on these motions of the pickleball paddle. The one or more parameters may be used to determine how frequently and how aggressively a pickleball paddle is used.

[0047] In an example, the one or more parameters may be based on a number of times the pickleball paddle strikes an object (e.g. pickleball). In an example, the one or more parameters may be based on a type/pattern of movements of the pickleball paddle. In an example, the one or more parameters may be based on an intensity of a movement by the pickleball paddle. The electronic component is connected to one or more sensors. The one or more sensors comprises at least one of: a movement sensor, a temperature sensor, a movement intensity sensor. The electronic component may comprise an electronic counter **320**. The electronic counter may be configured with a parameter based on a number of movements of the pickleball paddle. The electronic counter may be configured with a parameter based on intensity of movements of the pickleball paddle.

[0048] The electronic component may comprise a timer **318**. The timer enables determination of a parameter related to an age of the pickleball paddle. The electronic component may store a date based on when the pickleball paddle is manufactured. The one or more parameters stored in a memory may be retrieved by an external device using a wireless technology or a wireline technology. The one or more parameters enable determining at least one second parameter related to how the pickleball paddle was used. The one or more parameters enable determining at least one second parameter related to a quality of the pickleball paddle.

[0049] In some example embodiments, the one or more parameters stored in a memory may be retrieved by an external device using a wireless technology or a wireline technology (e.g., network interface **324**). The external device may comprise a process (e.g., FIG. 7) for determining performance of a pickleball paddle compared with when the pickleball paddle was new. In some embodiments, the process may at least in part be implemented in a processing circuitry **322** of the electronic component **312** in association with the external device. A user may enter a model and/or manufacturer of the pickleball paddle via a user interface. Different pickleball paddles may have different lifespans based on their manufacturing quality. The external device may determine a performance of the pickleball paddle based on the one or more parameters, model and/or manufacturer of the pickleball paddle, a date (e.g. year) the pickleball paddle was made, and/or other parameters related to the pickleball paddle. The external device may determine a factor that enables determining a performance of the pickleball paddle. For example, the performance of the pickleball paddle may be determined as a number between 1 to 10, a number between 0 to 5, and/or the like. For example, the performance of the pickleball paddle may be determined as low, medium or high. Other similar mechanisms may be implemented.

[0050] The electronic components may enable determining when it is time to change and/or discard a pickleball paddle as its performance has deteriorated and/or does not meet certain expectations. The electronic components may enable determining a value of a used pickleball paddle. Currently, there is no reliable mechanism to determine how frequently, how intensely and/or how long a used pickleball paddle is used. Example embodiments provide viable low-cost solutions to solve this issue.

[0051] In some example embodiments, the electronic component may be implemented with various motion, speed, motion direction, and acceleration sensing and memory. Various motions parameters may be recorded in a memory. These parameters may be transmitted to a processing tool for performing various parameters of playing styles and techniques of a pickleball players using the pickleball paddle. For example, the example embodiment may be configured to track pickleball paddle motion patterns during dinking, driving of a player using the pickleball paddle. A process may analyze the motion patterns and provide suggestions on how to improve the pickleball player's playing techniques and moves.

[0052] The electronics in the pickleball paddle may consume low electrical power. An initial battery and/or other sources **326** of electrical energy included with the electronic component may substantially survive until or after a pickleball paddle substantially is usable.

[0053] In some example embodiments, the pickleball paddle **400** may, in addition to an electronic component **412**, further comprise an electronic display **430**. The electronic component **412** may be connected to the electronic display **430**. The electronic component **412** may, similarly to electronic component **312**, include one or more sensors, memory, timer, counter, processing circuitry and power element. The display may indicate and/or display at least one indicator based on the one or more parameters. The electronic display may be a part of the striking blade (e.g., on the side frame) or a part of the handle. In an example, the electronic display may be an LED. The LED may display a color indicating how the pickleball paddle is used. For example, the color green may indicate the paddle usage was low. The color orange may indicate medium usage. The color red may indicate high usage. In an example, intensity of the LED light may indicate the pickleball paddle usage. Other similar examples may be implemented.

[0054] In an example, the electronic display may be a display bar. The display bar may indicate how the pickleball paddle is used. For example, the size of a bar on the display bar may indicate pickleball paddle usage, quality, and/or the like. The size may be numbered, for example, between 1 to 10. In an example, other display mechanisms may be used to indicate an indicator based on the one or more parameters. The indicator may indicate a usage and/or a quality of the paddle. The electronic display may be an LED, an LCD display, and/or other display mechanisms.

[0055] In some examples, the electronic display is turned on when one or more sensors are touched and/or activated. For example, the pickleball paddle may include two sensors **410** and **420**. When both sensors are touched and/or activated an electronic current may be initiated that causes the electronic display **430** to turn on and/or display an indicator. In some examples, only one touch sensor may be used. In an example, if the sensor is touched for a predetermined time duration longer than a threshold, the display may be turned on. The display is turned on when required to reduce electric energy usage by the pickleball paddle.

[0056] In some examples, the pickleball paddle may include a battery. In some examples, the pickleball paddle may include a solar panel, or other electronic components that produces electrical energy and voltage to charge the battery and/or other energy storage mechanisms. In an example, electrical energy may be generated based on movements of the pickleball paddle. In an example, there may be no need to change/replace the battery of a pickleball paddle. The electronics in the pickleball paddle may consume low electrical power. The initial battery and/or other sources of electrical energy may substantially survive until or after a pickleball paddle substantially is usable.

[0057] In an example, the electronic component, the sensors and/or the display may be a separate device that can be manufactured separate from a pickleball paddle, and can be attached to the pickleball paddle. A user may purchase the electronic component, the sensors and/or the display separately if desired. The user may connect them to the pickleball paddle to determine how the pickleball paddle is used and when it should be replaced. In this case, the one or more parameters may indicate parameters of the pickleball paddle based on when electronic devices are attached to the pickleball paddle.

[0058] Pickleball paddles have a limited lifespan depending on many factors as described in previous paragraphs.

Unfortunately, the certification testing process may not be adequate to determine whether a paddle is performing well or not. A damaged pickleball paddle may successfully pass certification testing. The test process for testing new pickleball paddles may not be applicable for testing a used pickleball paddle, as various defects introduced in used pickleball paddles are based on wear/tear, how the pickleball paddle was maintained and/or used, erosion, and many other factors that are not applicable to new pickleball paddles. A newly manufactured pickleball paddle is expected to perform according to manufacturing process throughout the striking blade of the pickleball paddle. However, wear and tear and damages on a paddle may be localized. For example, the entire surface of both sides of the striking blade may function as expected, except only one small part of a side of the striking blade that was subject to a damaging impact that is not visible. Example embodiments provide methods, apparatuses and systems for effectively testing a used pickleball paddle to determine its performance.

[0059] Some example embodiments implement a test process, apparatus, and system for testing a pickleball paddle. For example, the test process is applied to a used pickleball paddle to determine whether the pickleball paddle is damaged and/or has low quality. The test result may be used to decide whether to discard, replace or trade the pickleball paddle. A test apparatus and system, such as the test apparatuses shown in FIG. 5 and FIG. 6, may enable implementing the test process on a pickleball paddle.

[0060] FIG. 5 schematically shows a pickleball paddle tester **502** that is configured to perform a used pickleball paddle test according to some embodiment. The pickleball paddle tester **502** may include a memory **510**, one or more sensors **512**, a power source **514**, and processing circuitry **516**. In some embodiments, the pickleball paddle tester **502** may include a display **506** and a keyboard **508**. In some embodiments, a computer **504** may be connected to the pickleball paddle tester **502**. Instructions for the pickleball paddle tester **502** to perform a test process **700** on a pickleball paddle **520** may be stored in the memory **510** and/or the connected computer **504**.

[0061] In some example embodiment, a test process **700** for testing a pickleball paddle may comprise, at operation **702**, receiving one or more model parameters of a pickleball paddle. The outcome of the test, and thus the particular set of parameters and their values, may depend on the pickleball paddle model being tested. There are thousands of different pickleball paddle models in the market. For example, one or more model parameters may comprise a manufacturer name, a model number, a year, one or more paddle specifications (e.g. handle size, paddle shape, size, and/or the like).

[0062] Example embodiments may, at operation **704**, determine a plurality of first parameters of the pickleball paddle based on the one or more model parameters. The plurality of first parameters may indicate one or more performance parameters on each of a plurality of locations on each side of the striking blade of the pickleball paddle. Typically, both sides of a new pickleball paddle are expected to perform substantially the same. Performance parameters may be applicable to both sides of a new pickleball paddle. In an example the plurality of performance parameters may include a first plurality of deflection parameters, bouncing parameters, firmness parameters, friction parameters, smoothness parameters, roughness parameters, and/or structural integrity parameters. The performance parameters are

associated with different locations on the striking blade when the pickleball paddle is substantially new (e.g., during manufacturing, before significant use of the paddle). The plurality of first parameters may comprise deflection parameters of the pickleball paddle. In an example, the plurality of first deflection parameters are associated with the pickleball paddle when the pickleball paddle was in a substantially new condition.

[0063] The plurality of first parameters may be determined by the test device (e.g., pickleball paddle tester) **502**. In an example, a new pickleball paddle **520** may be placed on the test device. The test device **502** may perform corresponding measurements on different locations. These measurements may be stored in a memory (e.g., memory **510**) of the test device as reference measurement and used when a used pickleball paddle with the same model is tested by the test device **502**. In an example, the plurality of first parameters may be provided by the manufacturer of the pickleball paddle. To increase reliability of the measurement for the first parameters, multiple tests may be performed on a location on a pickleball paddle and/or on both sides of the striking blade. Example test locations **802** are shown in FIG. 8. FIG. 6 shows an example testing mechanism that can be used in the test device **502**. Average function or other functions on the measurement values may be used to determine the first plurality of parameters. In some examples, testing may be performed using different methods, for example, using different directional movements, different angles of the pickleball paddle position, or different temperatures.

[0064] In an example, the plurality of parameters may comprise a plurality of deflection parameters. A pickleball paddle deflection parameter may be based on and may determine how a pickleball paddle responds under force and this affects the game and a speed of the pickleball when it is hit by the pickleball paddle. A deflection parameter may be measured by various different mechanisms. In an example, deflection parameter at a first location of a striking blade may be related to how a pickleball bounces when the pickleball is hit by the first location on the pickleball paddle. To measure the deflection at a location of the pickleball paddle, the pickleball paddle may be positioned on the test tool. A standardized force may be applied and a response by the pickleball paddle may be measured. In an example, the deflection testing in the process may be different from paddle deflection testing during paddle certification. For example, a used pickleball paddle may be damaged at certain locations of the striking blade due to delamination. In the test process, deflection testing measurement may be performed at various locations of the striking blade. During a pickleball game, the pickleball may hit any location on the pickleball paddle. This test process may enable to determine how the striking blade of a used pickleball paddle is deteriorated at different locations (e.g., **802**) of different sides of a pickleball paddle.

[0065] In an example, the deflection parameter may be measured by simulating a pickleball hitting a striking blade of a pickleball paddle. This may be performed by an actual pickleball or some other objects simulating the role of a pickleball. For example, a force may direct an object (simulating a pickleball) to the pickleball paddle and measure how the object is returned (e.g. at what speed, with what force, with what acceleration, and/or the like). As shown in the example testing mechanism shown in FIG. 6, a simulated

pickleball **610** may be moved using a system of lines and pulleys (e.g., **606**, **608**, **612**, **614**) to move the simulated pickleball **610** over railings **602** and **604** to test the pickleball paddle **520** by impacting it at locations such as that shown in FIG. 8.

[0066] In an example, the plurality of first parameters may comprise a first plurality of spin parameters for example, a plurality of friction parameters, smoothness parameters, roughness parameter, grit parameters, and/or the like at different locations of the striking blade. In an example, many locations on both side of the striking blade may have the same value of spin parameters. In an example, these parameters may affect how well the pickleball spins a pickleball in a shot and/or how fast a pickleball ball returns when it hits the pickleball paddle. A pickleball paddle parameter may be based on and may determine how a pickleball paddle spins a pickleball and/or how fast it returns a pickleball, affecting dynamics of a game. A spin parameter may be measured by various different mechanisms. In an example, spin parameter at a first location of a striking blade may be related to how a pickleball spins when the pickleball is hit at the first location on the pickleball paddle at a first hitting motion direction. To measure the spin parameter at a location of the pickleball paddle, the pickleball paddle may be positioned on the test tool. A standardized object movement may be applied to the pickleball paddle and a response by the pickleball paddle is measured. In an example, the spin testing in the process may be different from paddle spin testing during paddle certification. For example, a used pickleball paddle may be damaged at certain locations of the striking blade due to wear and tear and erosion. In the test process, spin testing measurement may be performed at various locations of the striking blade. During a pickleball game, the pickleball may hit any location on the pickleball paddle. This test process may enable to determine how the striking blade of a used pickleball paddle is deteriorated at different locations of a side of a striking blade of a pickleball paddle.

[0067] In an example, surface roughness may affect a performance of pickleball paddles. In an example, roughness may be measured at different locations as a measure of how a pickleball paddle creates spin and/or speed. Roughness may be measured at different locations and/or different directions on the surface of both sides of a striking blade. In an example, a coefficient of friction may be measured at different locations on a surface of the striking blade. A kinetic coefficient of friction may indicate how a pickleball paddle grips a pickleball and this may affect spin and trajectory of the pickleball. The measurement may be performed at various locations of a pickleball paddle.

[0068] In an example, the spin parameter may be measured by simulating a pickleball hitting a striking blade of a pickleball paddle. This may be performed by an actual pickleball or some other object(s) simulating the role of a pickleball. For example, a force may direct an object (simulating a pickleball) to the pickleball paddle and measure how the object is spun or how much friction is generated. In an example, at least one smoothness/roughness factor may be measured by a test device measuring smoothness/roughness of the pickleball blade surface using mechanical or optical measurement techniques.

[0069] Example embodiments may comprise a database (e.g., stored in memory **510** and/or on the connected computer **504**) of measured parameters of a new pickleball

paddle for various pickleball models. In an example, the database may be implemented based on data provided by a manufacturer. In an example, the database may be updated based on measurements of new pickleball paddles of various models.

[0070] An example test device is shown in FIGS. 5 and 6.

[0071] In an example, at operation 706, the test device 502 may measure a plurality of second parameters on a first plurality of locations of a first side of (of the striking blade) the pickleball paddle. The plurality of the second parameters may be deflection parameters, spin parameters, a plurality of friction parameters, smoothness parameters, grit parameters, roughness parameters, and/or the like.

[0072] At operation 708, the test device 502 may measure a plurality of third parameters on a second plurality of locations of a second side of the pickleball paddle. The plurality of the third parameters may be deflection parameters, spin parameters, a plurality of friction parameters, smoothness parameters, grit parameters, roughness parameters, and/or the like.

[0073] At operation 710, the test device 502 may determine one or more fourth parameters indicating a quality of the pickleball paddle based on comparing the plurality of first parameters with a function of: the plurality of second parameters and the plurality of third parameters. The test device may send the one or more fourth parameters to a display (e.g., 506 and/or the display of computer 504). In an example, the pickleball paddle 520 is a used pickleball paddle. The quality of the pickleball paddle, based on the calculated fourth parameters, may be indicated as one of a plurality of predetermined values or value ranges.

[0074] The plurality of first parameters are measured on the pickleball paddle when the paddle is new. The plurality of first parameters comprises, for example: a plurality of fourth parameters of the first plurality of locations of the first side of the pickleball paddle; and a plurality of fifth parameters of the second plurality of locations of the second side of the pickleball paddle. A plurality of fourth parameters may be the same as the plurality of fifth parameters.

[0075] In an example, the test device may determine one or more fourth quality parameters indicating a quality of the pickleball paddle based on comparing: the plurality of fourth parameters with the plurality of second parameters; and the plurality of fifth parameters with the plurality of third parameters. The one or more fourth quality parameters is based on: at least one first parameter based on comparing the plurality of fourth parameters with the plurality of second parameters; and at least one second parameter based on comparing the plurality of fifth parameters with the plurality of third parameters. The one or more quality fourth parameters may be selected among: at least one first parameter; and at least one second parameter. The selection may be based on comparing the at least one first parameter with the at least one second parameter.

[0076] The process 700 shown in FIG. 7 is described above primarily in relation to the test device 502. However, it will be clear that the process is applicable to determining the quality of a pickleball paddle such as those described in relation to FIG. 3 or FIG. 4, and instead of or in addition to the parameters obtained from test device 502, use the parameters stored in the electronic components 312 or 412 in determining the quality.

[0077] FIG. 9 shows an example simplified test result and analysis. For simplicity, the test result is shown for one side

of the striking blade of the pickleball paddle. The difference in performance parameter of a used pickleball paddle and a new pickleball paddle is shown. The difference in terms of absolute value and percentage is shown in the last two columns. In an example, average value of the percentages or the absolute different may be used to determine a quality of the pickleball paddle.

[0078] In an example, a value in the measurement result may indicate that the pickleball paddle is damaged at one location. For example, at location 9, the performance of the paddle is degraded by 50%. In an example, if the performance of any of the locations is below a threshold, for example, 60%, the pickleball paddle may be discarded. In an example, the threshold may be different at different locations. In an example, different thresholds (e.g. for different locations) may indicate different quality indicators, e.g. high, medium, or low.

[0079] The quality indicator of the paddle may indicate the quality of the paddle according to various parameters. For example, the test result may indicate that the quality is medium for deflection testing, and the quality is low for spin testing. In an example, one quality factor may indicate the overall quality of the paddle based on a function of various testing parameters, e.g. an overall quality of a pickleball may be indicated as one of low, medium or high, or as a numeric value.

[0080] The above described embodiments provide efficient means by which used pickleball paddles can be evaluated, so that players can make decisions as to whether the pickleball paddles are still usable for playing pickleball without substantially affecting the games. It should be noted that embodiments of the present disclosure are not limited to the shapes, lengths or construction materials of the handles, strike blades, or butt caps of the pickleball paddles. Although various embodiments have been shown and described in detail, the claims are not limited to any particular embodiment or example.

1. A pickleball paddle, comprising
 - a striking blade;
 - a handle connected to the striking blade; and
 - an electronic component including a memory storing one or more parameters associated with how the pickleball paddle is used after the pickleball paddle is manufactured.
2. The pickleball paddle of claim 1, wherein the one or more parameters are based on a number of times the pickleball paddle strikes an object.
3. The pickleball paddle of claim 1, wherein the one or more parameters are based on a type of movement by the pickleball paddle.
4. The pickleball paddle of claim 1, wherein the one or more parameters are based on an intensity of a movement by the pickleball paddle.
5. The pickleball paddle of claim 1, wherein the electronic component is connected to or comprises one or more sensors, and wherein the one or more sensors comprise at least one of: a movement sensor, a temperature sensor, a movement intensity sensor.
6. The pickleball paddle of claim 1, wherein the electronic component comprises an electronic counter.
7. The pickleball paddle of claim 6, wherein the electronic counter is configured with a parameter based on a number of movements of the pickleball paddle.

8. The pickleball paddle of claim 7, wherein the electronic counter is configured with a parameter based on an intensity of movements of the pickleball paddle.

9. The pickleball paddle of claim 7, wherein the electronic component comprises a timer, wherein the timer enables determination of a parameter related to an age of the pickleball paddle.

10. The pickleball paddle of claim 7, wherein the one or more parameters are stored in the memory and configured to be retrieved by an external device using a wireless technology or a wireline technology.

11. The pickleball paddle of claim 7, wherein the one or more parameters enable determining at least one second parameter related to how the pickleball paddle was used and/or to a quality of the pickleball paddle.

12. A pickleball paddle, comprising
a striking blade;
a handle connected to the striking blade; and
an electronic component including a memory storing one or more parameters associated with how the pickleball paddle is used after the pickleball paddle is manufactured, wherein the electronic component is connected to an electronic display.

13. The pickleball paddle of claim 12, wherein the electronic display is an LED.

14. The pickleball paddle of claim 13, wherein the LED displays a color indicating how the pickleball paddle is used.

15. The pickleball paddle of claim 12, wherein the electronic display is a display bar, the display bar indicating how the pickleball paddle is used.

16. A test process for testing a pickleball paddle, comprising
receiving one or more model parameters of a pickleball paddle;
determining a plurality of first deflection parameters of the pickleball paddle based on the one or more model parameters, the plurality of first deflection parameters

being associated with the pickleball paddle when the pickleball paddle was in a substantially new condition;
measuring a plurality of second deflection parameters on a first plurality of locations of a first side of the pickleball paddle;
measuring a plurality of third deflection parameters on a second plurality of locations of a second side of the pickleball paddle;
determining one or more fourth parameters indicating a quality of the pickleball paddle based on comparing the plurality of first deflection parameters with a function of:
the plurality of second deflection parameters; and
the plurality of third deflection parameters; and
sending the determined one or more fourth parameters to a display.

17. The test process of claim 16, wherein the quality of the pickleball paddle is indicated as one of a plurality of predetermined values.

18. The test process of claim 16, wherein the plurality of first deflection parameters are measured on the pickleball paddle when the paddle was new.

19. The test process of claim 16, wherein the plurality of first deflection parameters comprises:

a plurality of fourth deflection parameters of the first plurality of locations of the first side of the pickleball paddle; and
a plurality of fifth deflection parameters of the second plurality of locations of the second side of the pickleball paddle.

20. The test process of claim 19, wherein the determining one or more fourth parameters indicating a quality of the pickleball paddle is based on comparing:
the plurality of fourth deflection parameters with the plurality of second deflection parameters; and
the plurality of fifth deflection parameters with the plurality of third deflection parameters.

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